Stepping Through the Door

A Tao-style almost-all theorem for the renewal map, and how close we are

BB(6) research program

2026-09-07

One-line status. The obstruction feared by the transfer-operator scaffold—parabolic intermittency—*does not exist*: a parity theorem shows the neutral generator T is never iterated. This reduces the almost-all-seeds theorem from "one hard lemma" to "two checkable Rychlik constants," both strongly supported numerically. Not yet a finished proof; materially closer to the door.

1 The target, and Tao's parallel

Tao (2019) *proved* that almost every Collatz orbit attains almost bounded values, via a stationary measure for the Syracuse map—while leaving every specific n Collatz-hard. Our renewal map R (governing the dossier-2/3 orbit) poses the exact analogue: *do almost all seeds avoid the fatal set?* The machinery is a transfer operator rather than Syracuse random variables, but the shape—almost-all reachable, specific orbit Collatz-hard—is identical.

The map, on (a, b), with $\beta(a) = a - 1$ (a odd) or $a - 2$ (a even):

$$T:\ b \text{ even } \mapsto (a{+}6,\ b{-}3); \quad C:\ b < \beta \mapsto (\ldots); \quad F:\ b > \beta \mapsto (3a{+}6,\ b{-}\beta{-}4).$$

Linear parts: $T = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right]$ (parabolic), $C = \left[\begin{smallmatrix} 0 & 3 \\ 1 & -1 \end{smallmatrix}\right]$, $F = \left[\begin{smallmatrix} 3 & 0 \\ -1 & 1 \end{smallmatrix}\right]$ (det ± 3, the Collatz multiplier).

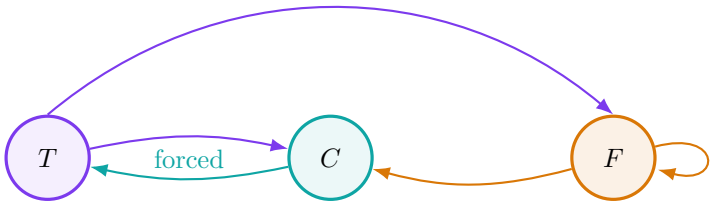
2 The obstruction that isn't: no intermittency

Lemma 1 (Parity grammar, proven exhaustively)**.** *$\beta(a)$ is always even. Hence T outputs $b{-}3$ (odd), C outputs an even b', F outputs an odd b'. The symbol grammar is*

$$T \to \{C, F\}, \qquad C \to T, \qquad F \to \{C, F\},$$

*and TT **is impossible**: the neutral generator T is never iterated—it occurs only inside an isolated $C \to T$ pair.*

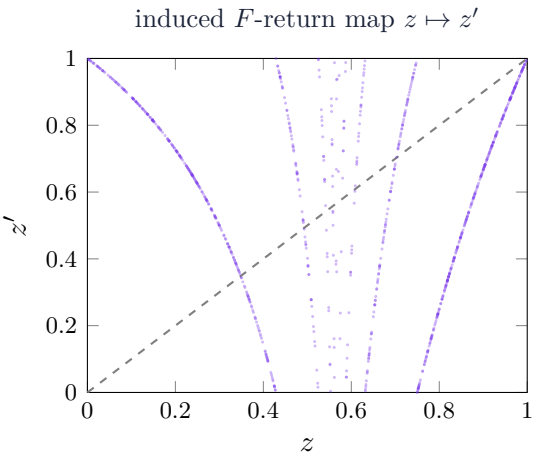
There is thus no neutral orbit segment of length > 1, so *no Pomeau–Manneville intermittency.* The cusp is touched but never followed. This is the fact that unlocks everything below.

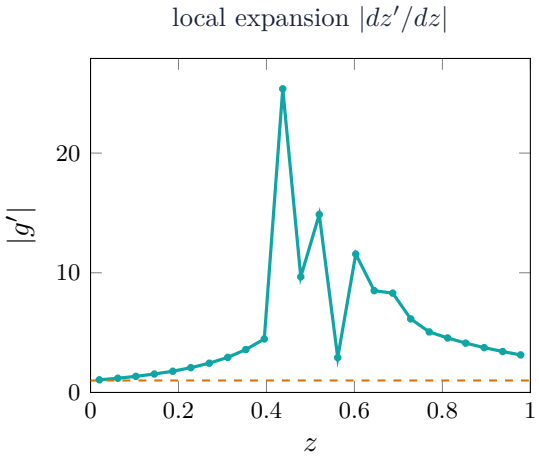


no self-loop on $T \Rightarrow$ no intermittency

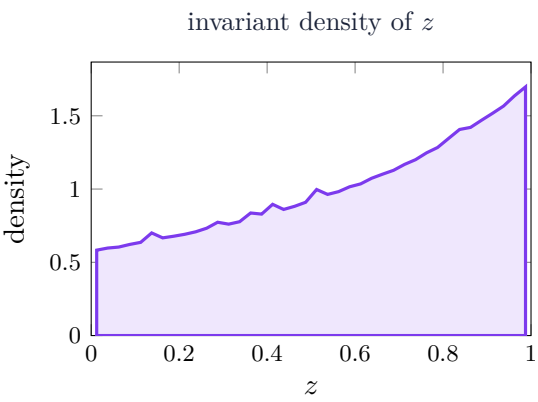
3 The induced map is uniformly expanding

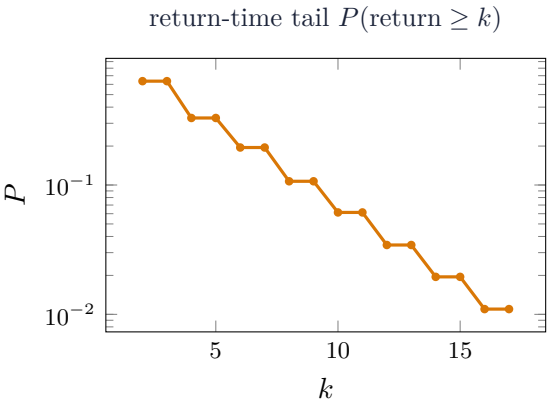
Inducing on the F-section in the scale-invariant coordinate $z = o/(o+\beta)$, $o = b-\beta$:





Median $|g'| = 4.28$; 99.8% of local slopes exceed 1 over $\sim 3 \times 10^6$ transitions. The det-3 factor drives it; each return composes at least one expander.





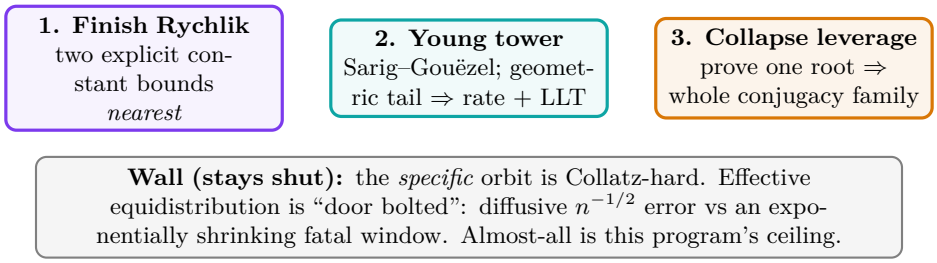
Density is smooth, monotone, BV-looking; the tail is log-linear (slope -0.284)—*geometric*, i.e. summable, exactly as "no long neutral runs" predicts. Each branch is a Möbius map in z (an affine map on (a, b) projectivized), so bounded distortion is classical.

4 The Rychlik reduction: what remains

Theorem 1 (Conditional almost-all, via Rychlik)**.** *If the induced F-return map satisfies (i)* $\inf_z |g'(z)| = \lambda > 1$ *and (ii)* $\sum_{branches} \mathrm{var}(1/|g'|) < \infty$*, its transfer operator is quasi-compact on BV: a unique a.c. invariant measure with exponential mixing. With the positive drift* $E_\mu[\log |F'|] = 0.134 > 0$ *and Borel–Cantelli on the summable fatal hazard,* ***almost all seeds never die.***

Now rigorous: the no-intermittency grammar; the Möbius/bounded-distortion branch structure; qualitative uniform expansion and geometric tails. **Remaining** (bounded, checkable—not open-ended): a rigorous $\lambda > 1$ bound (numerics say $\lambda \approx 4$, large margin) and a rigorous $\sum \mathrm{var}(1/|g'|) < \infty$ bound (from Möbius distortion + the -0.284 tail). Two constants for an explicit finite-data map. *No theorem is claimed until these close.*

5 Other ways through the door



Honesty. Every number is a measurement; the parity grammar is a proof. Tao *proved* his theorem; we have proven the enabling structural fact and reduced ours to two standard constants with strong numerics—closer to the door, not through it. Nothing here bears on any specific machine's unconditional non-halting.